When do local repairs of a P_3 packing combine into a cluster editing?

(working title; draft skeleton)

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Abstract

Li, Pilipczuk and Sorge study Cluster Editing parameterized by the excess ℓ of the solution size over a modification-disjoint packing H of induced P_3's. We ask which interaction structure of H makes this parameterization tractable, and use the treewidth w of the intersection graph $I(H)$ of the packing as the measure. We prove: (i) for $\ell = 0$ the problem is a CSP with domain size 3 whose constraint scopes are cliques of $I(H)$, hence solvable in $3^{w+1}n^{O(1)}$ given a tree decomposition; (ii) for every ℓ it is solvable in $7^{w+1}n^{2\ell+O(1)}$ given a tree decomposition (XP in ℓ, FPT in w), and domain size 7 cannot be lowered in the natural encoding; (iii) under ETH there is no $2^{o(\mathrm{tw}(I(H)))}n^{O(1)}$ algorithm even for $\ell = 0$ (exponent only, no base); (iv) w alone does not suffice (NP-hard for $w = 0$ with unbounded ℓ), and $\mathrm{tw}(G)$ and $\mathrm{tw}(I(H))$ are incomparable, with a tradeoff $\ell + \mathrm{tw}(I(H)) \geq 2m - 2$ on stars; (v) adjacent true twins lie in one cluster in every optimum, and if H is maximal among pair-disjoint packings (no induced P_3 has all three pairs unpacked), then in *every* optimum no cluster contains more than one whole twin class of vertices outside $V(H)$, so no edit has both endpoints outside $V(H)$; we derive a normal form (frozen classes, a shrink rule with a tight threshold, a size bound $|R| \leq (\ell+1)|V(H)|$, a classification of the partners of edits touching R) and show by a five-vertex example that pair-maximality cannot be weakened to maximality among vertex-disjoint packings; (vi) CEaMP is solvable in time $1.62^{(\binom{s}{2}+1)\ell}n^{O(1)} + n3^s$ and has a kernel with at most $(s^2-s+2)\ell$ vertices, where s is the largest number of vertices spanned by a connected component of $I(H)$; in particular $I(H)$ a perfect matching is FPT in ℓ. This is proved from a rule of van Bevern, Froese and Komusiewicz after verifying its hypotheses at the level of components; the parts that rest on their published proofs are listed explicitly. The main question, whether the problem is fixed-parameter tractable in $\ell + w$, is *open*; we do not prove or disprove it. (vii) For every packing H and every YES instance we prove counting lemmas on the structure of solutions: maximalising H by pair-disjoint induced P_3's costs at most $+|M| \leq \ell$ in treewidth (tight on stars); the hard non-adjacent pairs among the non-packed neighbours of a vertex have matching number at most ℓ, so the independence number of that neighbourhood is at most $2\ell + c + 1$; pieces of a solution that share a cluster are small ($|C| \leq \max\{D + u + 1, 2D, 2u\}$ for a cluster with at least two pieces); and the graph of packed pairs has treewidth at most $3w+2$. A clearly labelled heuristic discussion explains why hub and grid-tiling encodings do not give hardness; it is not a theorem. These lemmas give neither an FPT algorithm in $\ell + w$ nor hardness.

1 Introduction

Cluster Editing asks for the fewest edge insertions and deletions that turn a graph G into a disjoint union of cliques. Every induced P_3 must be hit by an edit, so a packing H of induced P_3's that pairwise share no vertex pair (a *modification-disjoint* packing) gives the lower bound $|H|$ on the

solution size. Li, Pilipczuk and Sorge [10] parameterize by the *excess* ℓ over this bound (problem CEaMP): is there a solution of size at most $|H| + \ell$? They show NP-hardness already for $\ell = 0$, with every vertex in at most 23 packed P_3's, and an $n^{2\ell+O(1)}$ algorithm when every vertex lies in at most two packed P_3's; whether the latter case is fixed-parameter tractable is open [10]. For vertex-disjoint packings the problem is FPT in ℓ [13].

The question of this paper is which measurable *interaction structure* of H makes the local repair choices (one edit per packed P_3, in three ways) combine into a global solution in $f(\ell, w)\, n^{O(1)}$ time. We take $w = \mathrm{tw}(I(H))$, where the intersection graph $I(H)$ has the packed P_3's as nodes and joins two of them when they share a vertex.

What is proved. For $\ell = 0$ the problem is a constraint satisfaction problem (CSP) with one variable of domain size 3 per packed P_3 and constraint scopes that are cliques of $I(H)$ (Section 3); dynamic programming over a tree decomposition gives $3^{w+1}n^{O(1)}$. For general ℓ, guessing the at most ℓ edits outside all packed P_3's reduces to the same CSP with domain size 7, giving $7^{w+1}n^{2\ell+O(1)}$ (Section 4). Section 5 proves a strict twin lemma and, for packings H that are maximal among pair-disjoint packings, the structure of the vertices R outside $V(H)$ and the statement (C1) that in every optimum no edit has both endpoints in R; it derives frozen classes, a shrink rule whose threshold is tight, the bound $|R| \le (\ell + 1)|V(H)|$, and a classification of the partners of edits touching R. A five-vertex example shows that the hypothesis cannot be weakened to maximality among vertex-disjoint packings, and a four-cycle shows that edits between R and $V(H)$ cannot be excluded. Section 6 proves Theorem K: with s the maximum number of vertices spanned by a component of $I(H)$, CEaMP runs in $1.62^{(\binom{s}{2}+1)\ell}n^{O(1)} + n3^s$ time and has a kernel with at most $(s^2-s+2)\ell$ vertices; $I(H)$ a perfect matching is therefore FPT in ℓ. The proof applies the reduction rule of [13] to the components of $I(H)$ after proving its hypotheses in full; it still relies on their published proof of the rule's soundness, on a Cluster Editing solver and on the $2k$-vertex kernel as quoted there. Section 7 proves counting lemmas for solutions with respect to an arbitrary packing (maximalisation with its cost in treewidth, a matching bound for hubs of deletions, small merges of pieces, and treewidth of the soft graph) and ends with a clearly labelled heuristic discussion of why simple hardness gadgets fail. Under ETH the exponent $\Theta(w)$ cannot be improved to $o(w)$ even at $\ell = 0$; w alone is insufficient; $\mathrm{tw}(G)$ and $\mathrm{tw}(I(H))$ are incomparable (Section 8).

What is not proved. Whether CEaMP is FPT in $\ell + \mathrm{tw}(I(H))$ (or W[1]-hard) is open, already for $w = 1$ with tree components of unbounded size. The algorithm of Section 4 is XP in ℓ, Theorem K needs bounded components, and no hardness is known. The normal form of Section 5 applies to a pair-maximal extension H' of the given packing, whose intersection graph may have larger treewidth, so it does not by itself give a preprocessing for the parameter $(\ell, \mathrm{tw}(I(H)))$. Section 9 states the open problems.

2 Preliminaries

Graphs are simple and undirected, $n = |V(G)|$. A *pair* is a 2-subset of $V(G)$; for a set S of pairs, $G \triangle S$ flips the adjacency of each pair in S. A graph is a *cluster graph* iff it has no induced P_3, i.e. no vertex triple with exactly two edges. A *solution* is a set S of pairs with $G \triangle S$ a cluster graph; $\mathrm{OPT}(G)$ is the minimum size of a solution.

A *packing* $H = \{P_1, \ldots, P_p\}$ is a set of induced P_3's $P_i = (a_i, b_i, c_i)$ ($a_ib_i, b_ic_i \in E$, $a_ic_i \notin E$). Write $\mathrm{pairs}(P_i)$ for the three pairs inside $V(P_i)$. H is *modification-disjoint* if the sets $\mathrm{pairs}(P_i)$ are pairwise disjoint; this is equivalent to: any two members share at most one vertex (two shared

vertices would be a shared pair, and any two vertices of a P_3 form one of its pairs). All packings below are modification-disjoint. A pair is *packed* if it lies in $\bigcup_i \text{pairs}(P_i)$ (its *owner* is then unique) and *unpacked* otherwise. For a vertex u, $Q(u)$ is the set of z with uz packed.

Problem CEaMP. Given G, a modification-disjoint packing H and $\ell \geq 0$, decide whether there is a solution S with $|S| \leq |H| + \ell$ (this is the convention of [10]; ℓ is the excess budget).

The *intersection graph* $I(H)$ has node set H and an edge PP' iff $V(P) \cap V(P') \neq \emptyset$. The P_3's through one vertex form a clique of $I(H)$. Hence (Helly property of tree decompositions) they lie in a common bag, and $c := \max_u |\{P \ni u\}| \leq \text{tw}(I(H)) + 1$ and $|Q(u)| \leq 2c$. We write $w = \text{tw}(I(H))$ with the convention $w = 0$ if $I(H)$ has no edge. A tree decomposition of width w is assumed *given* in the algorithmic statements; computing one costs $2^{O(w^3)}n$ exactly (Bodlaender [2]) or $2^{O(w)}n$ for a factor-2 approximation (Korhonen [9]) (the stated running times are as commonly quoted; we did not re-read them), which would change the bases below (e.g. 7^{w+1} to 7^{2w+2}).

Lemma 1 (Solutions decompose)**.** *Let S be any set of pairs such that $G \triangle S$ is a cluster graph, $x_i = S \cap \text{pairs}(P_i)$ and $U = S \setminus \bigcup_i \text{pairs}(P_i)$. Then every $x_i \neq \emptyset$, $|S| = \sum_i |x_i| + |U|$, and $|U| \leq |S| - |H|$.*

Proof. If $x_i = \emptyset$ the triple $V(P_i)$ keeps its adjacencies in $G \triangle S$ and is an induced P_3. The pair sets are disjoint, so $|S| = \sum |x_i| + |U| \geq |H| + |U|$. □

Remark 2 (Maximality). *H is pair-maximal* if no induced P_3 of G has all three pairs unpacked, i.e. if H is maximal among modification-disjoint packings. This is strictly stronger than maximality among vertex-disjoint packings (Proposition 15), and it is the only notion used in Section 5. If $H' \supseteq H$ adds t further pair-disjoint induced P_3's, then $|H'| + (\ell - t) = |H| + \ell$, so (G, H, ℓ) and $(G, H', \ell - t)$ ask the same question (a NO instance if $\ell < t$, since OPT $\geq |H'|$). A pair-maximal H' is found greedily in polynomial time. Caution: $\text{tw}(I(H'))$ may exceed $\text{tw}(I(H))$.

3 The case $\ell = 0$ is a CSP on the intersection graph

Observation 3. *Let S be a solution with $|S| \leq |H|$. Then S has exactly one pair in each* $\text{pairs}(P_i)$ *and no other pair, and $G \triangle S$ is a cluster graph iff no vertex triple is an induced P_3 of $G \triangle S$. Consequently $(G, H, 0)$ is a YES instance iff the following CSP is satisfiable: variables $x_i \in \{1, 2, 3\}$ (which pair of P_i is flipped), and for every vertex triple t the constraint "t is not an induced P_3 after flipping", whose truth depends on the owners' variables of the packed pairs of t only (unpacked pairs keep their status in G).*

Proof. By Lemma 1, $|x_i| \geq 1$ for all i and $|S| = \sum |x_i| + |U| \leq |H|$ force $|x_i| = 1$ and $U = \emptyset$. The rest is the definition of a cluster graph. □

The CSP has $\Theta(n^3)$ constraints, not $O(|H|)$.

Observation 4. *The set of owners of the packed pairs of any triple is a clique of $I(H)$ (at most three nodes). Hence the primal graph of the CSP is a subgraph of $I(H)$ (not every edge of $I(H)$ is witnessed by a triple), and a tree decomposition of $I(H)$ is one of the primal graph.*

Proof. Two pairs of one triple share a vertex v; their owners both contain v, so they coincide or are adjacent in $I(H)$. □

Theorem 5. *Given (G, H) and a tree decomposition of $I(H)$ of width w, CEaMP with $\ell = 0$ is solvable in $3^{w+1}n^{O(1)}$ time.*

Theorem 5 is the case $\ell = 0$ of Theorem 6 below (with a larger base there, 7, because for $\ell > 0$ a packed P_3 may receive several edits); its proof is the same dynamic program with domain $\{1, 2, 3\}$. Observation 4 is what makes the interaction of disjoint P_3's at $\ell = 0$ completely visible in $I(H)$: no transitivity-mediated interaction exists beyond shared vertices.

4 Excess edits: an XP-in-ℓ, FPT-in-w algorithm

Theorem 6. *Given (G, H, ℓ) and a tree decomposition of $I(H)$ of width w, CEaMP is solvable in $7^{w+1}n^{2\ell+O(1)}$ time.*

Proof. Let $\mathcal{D}_i$ be the 7 non-empty subsets of pairs(P_i). By Lemma 1, a solution of size $\leq |H| + \ell$ has the form $S = U \cup \bigcup x_i$ with $x_i \in \mathcal{D}_i$ and U a set of at most ℓ unpacked pairs; there are at most $\sum_{j \leq \ell} \binom{n^2/2}{j} \leq n^{2\ell}$ candidates for U.

Fix U and put $G_U = G \triangle U$. Since U avoids packed pairs, every P_i is still an induced P_3 of G_U with the same pairs, so H is still a modification-disjoint packing. For a triple t, the status of each unpacked pair of t in $G_U \triangle x$ is fixed, and each packed pair is determined by G_U and the variable of its owner. Hence t gives a constraint C_t ("t is not an induced P_3 in $G_U \triangle x$") whose scope is the owner set of t, a clique of $I(H)$ by Observation 4 (the proof there does not use G itself). Triples with empty scope are tested directly, in $O(n^3)$ time. Seek $x \in \prod \mathcal{D}_i$ satisfying all C_t and minimizing the unary cost $\sum_i |x_i|$.

Let $(T, \{B_s\})$ be the tree decomposition. By the Helly property each clique lies in a bag; assign each constraint to one such bag. For a node s and an assignment α of B_s, let $f_s(\alpha)$ be the minimum cost of $\sum |x_i|$ over variables in the subtree of s, among assignments extending α and satisfying all constraints assigned in the subtree. The standard recurrences (introduce, forget, join; see [4]) compute all f_s in time $7^{w+1}n^{O(1)}$ since there are $O(n^3)$ constraints and each table has $\leq 7^{w+1}$ entries. Accept iff for some U the minimum m_U satisfies $m_U + |U| \leq |H| + \ell$. Soundness: any feasible (U, x) gives the solution $U \cup \bigcup x_i$. Completeness: Lemma 1. □

Remark 7 (Why domain size 7). The empty value is never feasible, so 7 (not 8) suffices. It cannot be lowered to 3: an edited packed P_3 can receive two or three edits in every optimum. For two: $n = 5$, $E = \{02, 04, 12, 23, 24\}$, $H = \{(1, 2, 3)\}$, OPT $= 2$, unique optimum $\{12, 23\}$. For three: $n = 6$, $E = \{01, 02, 03, 04, 12, 14, 15, 24, 25, 35, 45\}$, $H = \{(0, 3, 5)\}$, OPT $= 3$, unique optimum $\{03, 05, 35\}$ (both checked exhaustively over all set partitions). In the notes' random data (computer evidence, $n \leq 9$) about 8–9% of instances force ≥ 2 edits on some packed P_3 and $\approx 0.2\%$ force three. For $c = 2$, $I(H)$ is subcubic but may have unbounded treewidth, so Theorem 6 is a different tradeoff from, not a recovery of, the $n^{2\ell}$ algorithm of [10].

Proposition 8 (Completing a fixed packed assignment)**.** *Let x assign a subset $x_i \subseteq$ pairs(P_i) to each P_i and let $B = |H| + \ell - \sum |x_i|$ be the excess budget left. Whether there is a set U of at most B unpacked pairs with $G \triangle (x \cup U)$ a cluster graph is decidable in $3^B n^{O(1)}$ time.*

Proof. Search tree on $G_x = G \triangle x$: if the current graph has an induced P_3, some pair of it must be flipped; packed pairs are frozen by x, so branch on its at most three unpacked pairs, reducing B by one (if all its pairs are packed, reject). Depth is at most B; flipping a pair twice is useless, so the search is exhaustive. □

Remark 9. Proposition 8 is a reformulation, not progress towards FPT in (ℓ, w): finding x is the whole problem, and the property "x is completable" is not local in x.

5 Twins, vertices outside the packing, and a normal form

By Lemma 1 the whole difficulty of $\ell > 0$ is the set U of at most ℓ unpacked edits. This section proves what can be said about U without any further assumption on the instance beyond maximality of H (in the pair sense of Remark 2).

Conventions. A *partition* Π of V into *clusters* has *cost* equal to the number of its *edits*: edges between different clusters (deletions) plus non-edges inside a cluster (insertions). Solutions S and partitions correspond ($G \triangle S$ is a cluster graph whose cliques are the clusters), so $\mathrm{OPT}(G)$ is the minimum cost and an *optimum* is a partition of minimum cost. $\Pi(v)$ is the cluster of v. Unless said otherwise, H is *pair-maximal* (Remark 2: no induced P_3 of G has all three pairs unpacked) and $R = V \setminus V(H)$; every pair with an end in R is unpacked.

5.1 Twins

Lemma 10 (Twins, strict form)**.** *Let $u \neq v$ be adjacent with $N[u] = N[v]$. For every partition Π with $\Pi(u) \neq \Pi(v)$, one of the two partitions obtained by moving u into $\Pi(v)$, respectively v into $\Pi(u)$, has cost smaller than Π by at least 1; more precisely the two cost changes sum to exactly -2. Hence u, v lie in one cluster in* every *optimum, with no hypothesis on H.*

Proof. For a vertex y and a set $D \not\ni y$ let $f(y, D) = |D \setminus N(y)| - |D \cap N(y)|$. Put $D_u = \Pi(u) \setminus \{u\}$ and $D_v = \Pi(v) \setminus \{v\}$ (possibly empty); $D_u \not\ni v$, $D_v \not\ni u$, and neither contains u or v. Moving u out of $\Pi(u)$ changes the status of the pairs uz, $z \in D_u$: before, uz is an edit iff it is a non-edge; after, iff it is an edge; the change is $-f(u, D_u)$. Moving u into $\Pi(v) = D_v \cup \{v\}$ changes, for $z \in D_v \cup \{v\}$, the status from "edit iff edge" to "edit iff non-edge"; the change is $f(u, D_v \cup \{v\}) = f(u, D_v) - 1$ because $uv \in E$. All other pairs keep their status. So

$$\Delta_1 = -f(u, D_u) + f(u, D_v) - 1, \qquad \Delta_2 = -f(v, D_v) + f(v, D_u) - 1.$$

Since $N(u) \setminus \{v\} = N(v) \setminus \{u\}$ we have $f(u, D) = f(v, D)$ for all $D \subseteq V \setminus \{u, v\}$, hence $\Delta_1 + \Delta_2 = -2$. Thus $\min(\Delta_1, \Delta_2) \leq -1$. □

This is the critical-clique property used in kernelizations of Cluster Editing (Guo [6]; also stated in [13]); the argument above is self-contained and shows that it is strict.

5.2 Viability of an unpacked edit

Lemma 11 (Viability of an unpacked edit)**.** *Let S be any solution, U its unpacked part and $uv \in U$. Let $D(uv)$ be the set of $z \notin Q(u) \cup Q(v) \cup \{u, v\}$ such that $\{u, v, z\}$ is an induced P_3 of $G \triangle \{uv\}$. Then for every $z \in D(uv)$ one of uz, vz lies in U, and $|D(uv)| \leq |U| - 1$.*

Proof. All three pairs of $\{u, v, z\}$ are unpacked ($z \notin Q(u) \cup Q(v)$). If neither uz nor vz is in S, the triple has in $G \triangle S$ the same pattern as in $G \triangle \{uv\}$, an induced P_3. Distinct z give distinct pairs, all different from uv. □

This gives a kernel-style condition (deleting uv needs fewer than ℓ common neighbours outside $Q(u) \cup Q(v)$) but no bound on the number of candidate pairs: all pendant edges are viable.

5.3 The vertices outside $V(H)$

Lemma 12 (Structure of R)**.** *Let H be pair-maximal. For $r, s \in R$: if $rs \in E$ then $N[r] = N[s]$; if $rs \notin E$ then $N(r) \cap N(s) = \emptyset$. Consequently:*

1. *every component X of $G[R]$ (a* class*) is a clique all of whose vertices have the same closed neighbourhood $N[X]$ (they are true twins);*
2. *$A_X := N(X) \setminus X$ is contained in $V(H)$;*
3. *the sets A_X, over the classes X, are pairwise disjoint.*

Proof. Every pair with an end in R is unpacked. Let $rs \in E$ and $z \neq r, s$ be adjacent to exactly one of r, s. Then $\{r, s, z\}$ has exactly two edges, i.e. is an induced P_3 with all three pairs unpacked, contradicting pair-maximality; so $N[r] = N[s]$. If $rs \notin E$ and z is a common neighbour, then $r\,z\,s$ is an induced P_3 whose pairs rz, zs, rs all have an end in R, again a contradiction.

(1) Adjacent vertices of R have equal closed neighbourhoods; along a connected component all closed neighbourhoods coincide, and then any two vertices r, t of the component satisfy $r \in N[r] = N[t]$, so they are adjacent. (2) A neighbour of $x \in X$ lying in R is in the component of x, hence in X. (3) If $a \in A_X \cap A_Y$ for classes $X \neq Y$, then a is a common neighbour of some $x \in X$ and $y \in Y$, which are non-adjacent (different components of $G[R]$); this contradicts the second clause. □

For $x \in X$ and $a \in V(H)$ we thus have $xa \in E$ iff $a \in A_X$; in particular vertices of one class are twins towards $V(H)$ as well. (Lemma 12 fails for vertex-maximal packings, see Proposition 15.)

Lemma 13 (Exchange formula)**.** *Let Π be any partition, X a class with $X \subseteq C$ for a cluster C, $x = |X|$ and $a = |A_X \cap C|$. Let Π' be obtained by moving X out of C into a cluster of its own. Then* $\mathrm{cost}(\Pi') - \mathrm{cost}(\Pi) = x\,(x + 2a - |C|)$.

Proof. Only the pairs between X and $C \setminus X$ change status. By Lemma 12, a vertex of X has exactly a neighbours in $C \setminus X$, namely $A_X \cap C$. Before the move such a pair is an edit iff it is a non-edge: $x(|C| - x - a)$ edits. After the move it is an edit iff it is an edge: xa edits. The difference is $xa - x(|C| - x - a) = x(x + 2a - |C|)$. □

Theorem 14 (C1: no edit inside R)**.** *Let H be pair-maximal. In* every *optimum Π, every cluster meets R in the empty set or in exactly one whole class. Hence no edit of Π has both endpoints in R, and for every class X the cluster $\Pi(X)$ equals $X \cup B_X$ with $B_X := \Pi(X) \cap V(H)$.*

Proof. By Lemma 12(1) and Lemma 10, every class lies inside one cluster (a class of size 1 trivially). Suppose a cluster C contains two distinct classes X_1, X_2. Put $x_i = |X_i| \geq 1$ and $a_i = |A_{X_i} \cap C|$. The sets $X_1, X_2 \subseteq R$ and $A_{X_1} \cap C, A_{X_2} \cap C \subseteq V(H)$ are pairwise disjoint (Lemma 12(2),(3)), so $|C| \geq x_1 + x_2 + a_1 + a_2$. Let Δ_i be the cost change of moving X_i out of C. Since Π is an optimum, $\Delta_1 \geq 0$ and $\Delta_2 \geq 0$, i.e. by Lemma 13 $|C| \leq x_i + 2a_i$ for $i = 1, 2$. With the lower bound on $|C|$: $x_2 + a_2 \leq a_1$ and $x_1 + a_1 \leq a_2$. Adding gives $x_1 + x_2 \leq 0$, absurd. So no cluster contains two classes.

An edit with both ends in R joins two vertices r, s. If they lie in one class, they are adjacent and (by the above) in one cluster, so rs is no edit. If they lie in different classes they are non-adjacent, and rs is an edit iff they share a cluster, which was excluded. □

The strictness of the inequality is what makes the statement hold for every optimum (no ties, and no iteration or termination argument).

Proposition 15 (Pair-maximality cannot be replaced by vertex-maximality)**.** *Let G have vertices $0,\dots,4$ and edges $03, 13, 14, 24, 34$, and $H = \{(0,3,1)\}$ (pairs $03, 13, 01$). Then H is maximal among vertex-disjoint packings (only the two vertices $2, 4$ remain), $R = \{2,4\}$, $\mathrm{OPT} = 2$ and the unique optimum is $\{03, 24\}$; it contains the edit 24 with both endpoints in R. So Theorem 14 fails. Here $\{3,4,2\}$ is an induced P_3 all of whose pairs are unpacked, so H is not pair-maximal and Lemma 12 fails.*

Proof. The partition $\{\{0\},\{1,3,4\},\{2\}\}$ has the two edits $03, 24$. Conversely, $0\,3\,1$ and $3\,4\,2$ are induced P_3's with disjoint pair sets $\{03, 13, 01\}$ and $\{34, 24, 23\}$, so $\mathrm{OPT} \geq 2$. That this is the only optimum was checked by enumerating all 52 partitions of the five vertices (computer; not re-derived by hand). □

5.4 Consequences

Let (G, H, ℓ) be a *YES instance* with H pair-maximal, Π any optimum, S its edit set, so $|S| = \mathrm{OPT} \leq |H| + \ell$. Let U be the unpacked edits. By Lemma 1, $|S| - |H| = \sum_i (|x_i| - 1) + |U|$, so $|U| \leq \ell$. Write $q = \max_u |Q(u)| \leq 2c$.

Corollary 16 (Normal form)**.** *For every class X let $B_X = \Pi(X) \cap V(H)$.*

(N1) Types of unpacked edits. *Every unpacked edit is either an* HH-edit *(both ends in $V(H)$) or an* RH-edit *$\{x, a\}$, $x \in X$ a class, $a \in V(H)$; the latter is a deletion if $a \in A_X \setminus B_X$ and an insertion if $a \in B_X \setminus A_X$, and $\{x, a\}$ is edited iff $\{x', a\}$ is, for all $x, x' \in X$. With h the number of HH-edits, $|U| = h + \sum_X |X| \cdot |A_X \triangle B_X| \leq \ell - \sum_i (|x_i| - 1) \leq \ell$.*

(N2) Frozen classes. *If $|X| > \ell$, then $\Pi(X) = X \cup A_X$.*

(N3) Attachment. *Every $a \in V(H)$ lies in A_X for at most one class X. If $z, z' \in A_X$ are non-adjacent then zz' is packed. If $a \in V(H) \setminus A_X$, $z \in A_X$ and $az \in E$ then az is packed; hence $|N(a) \cap A_X| \leq |Q(a)| \leq q$.*

(N4) Shrink rule. *If $|X| \geq \ell + 2$ and $x \in X$, then (G, H, ℓ) is a YES instance iff $(G - x, H, \ell)$ is. This holds for H pair-maximal without the YES hypothesis. If $A_X = \emptyset$ then X is a component of G and may be deleted without changing* OPT*.*

(N5) Size. *After exhausting the two rules in (N4), every class has at most $\ell + 1$ vertices and at most $|V(H)|$ classes have $A_X \neq \emptyset$, so $|R| \leq (\ell + 1)|V(H)|$ and $n \leq (\ell + 2)|V(H)| \leq 3(\ell + 2)|H|$.*

Proof. (N1) Pairs with an end in R are unpacked; by Theorem 14 none has both ends in R. The vertices of X are twins towards $V(H)$ and lie in one cluster, which gives the all-or-none statement; xa is an edit iff exactly one of $a \in B_X$, $a \in A_X$ holds. Counting gives the identity; the bound is the first paragraph of this subsection.

(N2) If $\Pi(X) \neq X \cup A_X$ then, since $\Pi(X) = X \cup B_X$ (Theorem 14), $A_X \triangle B_X \neq \emptyset$ and (N1) gives $|U| \geq |X| > \ell$, contradicting $|U| \leq \ell$.

(N3) The first claim is Lemma 12(3). Let $x \in X$. If $z, z' \in A_X$ are non-adjacent, $z\,x\,z'$ is an induced P_3 with pairs zx, xz' unpacked, so zz' must be packed (pair-maximality). If $a \in V(H) \setminus A_X$ and $z \in A_X$ with $az \in E$, then $xa \notin E$ and $x\,z\,a$ is an induced P_3 with xz, xa unpacked, so az is packed, i.e. $z \in Q(a)$.

(N4) $x \in R$, so H is still a packing of $G - x$, still pair-maximal (an induced P_3 of $G - x$ is one of G), and $X \setminus \{x\}$ is a class of $G - x$ with the same A_X. ($\Rightarrow$) Restricting a partition of G to $V \setminus x$ does not increase its cost, so $\mathrm{OPT}(G - x) \leq \mathrm{OPT}(G) \leq |H| + \ell$. ($\Leftarrow$) Let Π be an optimum of $G - x$

with cost $\leq |H| + \ell$. Now $|X \setminus x| \geq \ell + 1 > \ell$, so (N2) in $G - x$ gives $\Pi(X \setminus x) = (X \setminus x) \cup A_X$. Add x to this cluster: $N[x] = X \cup A_X$, so every pair at x is satisfied, and the cost does not change. Hence $\mathrm{OPT}(G) \leq |H| + \ell$. If $A_X = \emptyset$, X is a clique component of G, which costs nothing, and H is disjoint from it.

(N5) After the rules each class has at most $\ell + 1$ vertices; the non-empty sets $A_X \subseteq V(H)$ are pairwise disjoint (Lemma 12), and $|V(H)| \leq 3|H|$. Classes with $A_X = \emptyset$ were deleted. □

Remark 17 (Tightness of the threshold in (N4)). For $\ell = 0$ the threshold $|X| \geq 2$ cannot be lowered to $|X| \geq 1$: take the 4-cycle of Proposition 20 below, $H = \{(0, 1, 3)\}$, $X = \{2\}$. Then $(G, H, 0)$ is a NO instance, but $G - 2$ is the induced P_3 0 1 3 itself, so $(G - 2, H, 0)$ is a YES instance. (For larger ℓ the notes report computer evidence of the same tightness; no proof is given here.)

Remark 18 (The rules and the user-supplied packing). The input packing H of CEaMP need not be pair-maximal. Extend it greedily (polynomial time) to a pair-maximal $H' \supseteq H$ with $t = |H'| - |H|$ new members, set $\ell' = \ell - t$ (answer NO if $\ell' < 0$, Remark 2) and apply this section to (G, H', ℓ'). Then (N4), (N5) hold with ℓ' and H'; in particular $n \leq 3(\ell + 2)(|H| + \ell)$ after reduction, since $|H'| \leq |H| + \ell$. But $\mathrm{tw}(I(H'))$ may exceed $\mathrm{tw}(I(H))$, so these rules are a preprocessing for the parameters $(\ell, |H'|)$ or $(\ell, \mathrm{tw}(I(H')))$, not for $(\ell, \mathrm{tw}(I(H)))$ with the given H; and $|V(H')|$ is not bounded by ℓ, so (N5) is not a kernel.

Proposition 19 (Partners of edits touching R)**.** *In the setting of Corollary 16, let X be a class, $B = B_X$. Then:*

(P1) Deletion. *If $\{x, a\}$ is an RH-deletion, $a \in A_X$, then X is determined by a (the unique class with $a \in A_X$): it is a unary flag at a of weight $|X|$. Moreover a is then cut off from every $z \in A_X \cap B$ with $z \notin Q(a)$, each such edge az being an unpacked HH-deletion; so at least $|A_X \cap B| - q$ HH-edits accompany it.*

(P2) No detachment. *If $B \cap A_X = \emptyset$ then $B = \emptyset$.*

(P3) Insertion. *If $a \in B \setminus A_X$ then $B \cap A_X \neq \emptyset$, and for every $z \in B \cap A_X$ either az is packed ($a \in Q(z)$, at most q choices of a per z) or az is an unpacked non-edge, hence an inserted HH-edit.*

(P4) Wide classes. *If $|A_X| \geq \ell + q$ then $B \subseteq A_X$.*

Consequently, every candidate partner $a \in V(H)$ of a class in an RH-edit is unary (P1), local to some z (P3, packed case) or tied to an HH-edit (P3, second case; P1). This classifies the partners; it does not assert that the HH-edits are independent of the RH-edits.

Proof. (P1) Uniqueness is (N3). Let $a \in A_X \setminus B$ and $z \in A_X \cap B$, $z \notin Q(a)$. If $az \notin E$ then az would be packed by (N3), contradicting $z \notin Q(a)$; so $az \in E$, unpacked, and $a \notin \Pi(X) \ni z$, so az is deleted. At most q vertices z lie in $Q(a)$.

(P2) Suppose $B \neq \emptyset = B \cap A_X$. The cluster is $C = X \cup B$ (Theorem 14), $a_X = |A_X \cap C| = 0$, and by Lemma 13 moving X out changes the cost by $|X|(|X| - |X| - |B|) = -|X||B| < 0$, contradicting optimality.

(P3) $B \neq \emptyset$, so $B \cap A_X \neq \emptyset$ by (P2). Let $z \in B \cap A_X$. If $az \in E$ then az is packed by (N3) ($a \in V(H) \setminus A_X$), i.e. $a \in Q(z)$. Otherwise either az is packed (again $a \in Q(z)$) or it is an unpacked non-edge inside the cluster $\Pi(X)$, an insertion.

(P4) Suppose $B \not\subseteq A_X$. Put $\alpha = |A_X \cap B|$, $\delta = |A_X \setminus B|$, $\beta = |B \setminus A_X| \geq 1$. By (N1) there are $|X|\beta$ RH-insertions and $|X|\delta$ RH-deletions. For $b \in B \setminus A_X$ and $z \in A_X \cap B$ with $z \notin Q(b)$, the

pair bz is unpacked, lies in a cluster, and is a non-edge by (N3); so it is an unpacked insertion, an HH-edit. There are at least $\beta \max(0, \alpha - q)$ of them. Hence $\ell \geq |U| \geq \beta(|X| + \max(0, \alpha - q)) + |X|\delta$. If $\alpha \geq q$ this gives $\ell \geq 1 + (\alpha - q) + \delta$, i.e. $|A_X| = \alpha + \delta \leq \ell + q - 1$. If $\alpha < q$ it gives $\delta \leq \ell - 1$ and $|A_X| \leq q - 1 + \ell - 1$. Either way $|A_X| < \ell + q$. □

5.5 Where locality stops

Proposition 20 (Strong locality fails)**.** *Let $G = C_4$ with edges $01, 02, 13, 23$ and $H = \{(0, 1, 3)\}$ (pairs $01, 13, 03$; vertex $2 \notin V(H)$). Then H is pair-maximal,* $\mathrm{OPT} = 2 = |H| + 1$*, so $(G, H, 0)$ is a NO instance, and there are exactly three optima $\{03, 12\}$, $\{01, 23\}$, $\{02, 13\}$, each containing an unpacked pair at vertex 2.*

This (checked exhaustively over all partitions, and verified by hand: every induced P_3 of C_4 contains one of the packed pairs $01, 13, 03$) refutes the claim that all unpacked edits of some optimum have both endpoints in $V(H)$; Theorem 14 is what remains true: no edit has both endpoints in R. In the notes' random sample ($n \leq 9$) about 4% of maximal-packing instances with $|R| \geq 2$ force an edit touching R (computer evidence, not a theorem).

Remark 21 (Vertex-disjoint packings and the case $c = 1$). For a vertex-disjoint H ($w = 0$, $c = 1$) FPT in ℓ is the theorem of [13]. One must not derive it from Theorem 14 by treating H as pair-maximal: a vertex-disjoint H that is maximal among vertex-disjoint packings need not be pair-maximal (Proposition 15; also the path $0\,1\,2\,3\,4$ with $H = \{(0, 1, 2)\}$, which is extended by $(2, 3, 4)$ sharing the vertex 2). The pair-maximal extension H' of Remark 18 is in general no longer vertex-disjoint, so $I(H')$ has edges and the case $w = 0$ is not available for it. We therefore claim nothing about $c = 1$ beyond [13] and Theorem 22.

6 Bounded components of $I(H)$: Theorem K

Let $C_1, \ldots, C_r$ be the connected components of $I(H)$, $p_j = |C_j|$ the number of P_3's in C_j, and $W_j = \bigcup_{P \in C_j} V(P)$. Put $s = \max_j |W_j|$, the largest number of *vertices* spanned by a component. If $s_P = \max_j p_j$ counts P_3's instead, then $s \leq 2s_P + 1$: order the P_3's of a component so that each shares a vertex with an earlier one; the first spans 3 vertices and each later one adds at most 2.

Theorem 22 (Theorem K)**.** *CEaMP can be solved in time $1.62^{(\binom{s}{2}+1)\ell} n^{O(1)} + n\, 3^s$ and admits a kernel with at most $(s^2 - s + 2)\,\ell$ vertices, where s is the maximum number of vertices spanned by a connected component of $I(H)$. (No hypothesis of maximality is used. The kernel is an instance of CEaMP with empty packing.)*

Corollary 23. *If $I(H)$ is a perfect matching (so $w = 1$, $c = 2$), CEaMP is FPT in ℓ: here $s = 5$, the time is $1.62^{11\ell} n^{O(1)}$ and the kernel has at most 22ℓ vertices. More generally components with at most s_P P_3's give Theorem 22 with $s = 2s_P + 1$.*

Proof of Corollary. Two adjacent nodes of $I(H)$ share exactly one vertex (H is modification-disjoint), so $|W_j| = 3 + 3 - 1 = 5$; $s^2 - s + 2 = 22$ and $\binom{5}{2} + 1 = 11$. □

What is cited and what is proved. We use the problem of van Bevern, Froese and Komusiewicz [13]: an instance $(G, \mathcal{G}, k)$ consists of a graph, a set $\mathcal{G}$ of *vertex-disjoint induced subgraphs* G_j with $1 \leq \tau(G_j) \leq t$, where τ is the cluster-editing optimum, and a budget k; its parameters are t and $\ell_B = k - \sum_j \tau(G_j)$ (their Def. 1.2, Problem 1.3, read in the arXiv text; connectivity of the

G_j is not required, and their size enters only through t). We prove below, in full, that the components of $I(H)$ define such an instance and that $\ell_B \le \ell$ (Lemma 24), that the reduction rule can be applied at component level and does not disturb the other components (Lemma 25), and the bound $r \le 2\ell_B$ on the number of irreducible components (Lemma 26; this re-derives, for components, the counting argument of BFK Lemma 3.2 and Lemma 6.5 instead of citing it). *The following rest on the published proofs of [13] and are not reproved here:* (i) soundness of their Reduction Rule 6.2 for an arbitrary induced subgraph $G[W]$ (their Lemma 6.3, whose proof uses only G, W and the optimal solution S of $G[W]$); (ii) the time to test and apply the rule (their Lemma 6.4); (iii) the algorithm of Böcker running in $O(1.62^k + n + m)$ and the kernel with at most $2k$ vertices for Cluster Editing, both as quoted in [13] (we have not opened the original sources).

Lemma 24 (The components form a packing in the sense of [13]). *Let $G_j = G[W_j]$, $\tau_j = \mathrm{OPT}(G_j)$, $k = |H| + \ell$ and $\ell' = k - \sum_j \tau_j$.*

1. *The sets W_j are pairwise disjoint and each G_j is an induced subgraph.*
2. *$p_j \le \tau_j \le \lfloor \binom{s}{2}/2 \rfloor =: t_s$ and $\tau_j \ge 1$.*
3. *$\ell' = \ell - \sum_j (\tau_j - p_j) \le \ell$.*
4. *For every solution S of G and every j, $S_j := S \cap \binom{W_j}{2}$ is a solution of G_j, so $|S_j| \ge \tau_j$; the sets $\binom{W_j}{2}$ are pairwise disjoint, so $|S| \ge \sum_j \tau_j + |S_{\mathrm{ext}}|$ where S_{ext} are the pairs of S inside no W_j. In particular $\mathrm{OPT}(G) \ge \sum_j \tau_j$, and if $\ell' < 0$ then (G, H, ℓ) is a NO instance.*

Proof. (1) A vertex in $W_i \cap W_j$, $i \ne j$, lies in a P_3 of C_i and in a P_3 of C_j; these are adjacent in $I(H)$, contradicting that C_i, C_j are different components. Induced by definition.

(2) The P_3's of C_j are induced P_3's of G_j with pairwise disjoint pair sets inside $\binom{W_j}{2}$. A solution of G_j must edit a pair of each of them (otherwise the triple survives as an induced P_3, as in Lemma 1); hence $\tau_j \ge p_j \ge 1$. For the upper bound, let m_j be the number of edges of G_j; deleting all edges, or inserting all non-edges, yields a cluster graph, so $\tau_j \le \min(m_j, \binom{|W_j|}{2} - m_j) \le \binom{|W_j|}{2}/2 \le \binom{s}{2}/2$.

(3) $|H| = \sum_j p_j$ because the components partition H.

(4) $(G \triangle S)[W_j] = G_j \triangle S_j$ is an induced subgraph of a cluster graph, hence a cluster graph. The rest is counting with disjoint pair sets. □

Thus (G, H, ℓ) is a YES instance iff $(G, \{G_j\}, k)$ is a YES instance of the problem of [13] with $t \le t_s$ and excess $\ell' \le \ell$ (answer NO if $\ell' < 0$). Note why components, not single P_3's: a pair uv with $u \in W_j$ and $v \notin W_j$ lies in no W_i at all when v is in R or in another component, hence is not counted in $\sum_j \tau_j$ and is an excess edit (Lemma 24(4)); for a packing whose members share vertices, the same pair may be a packed pair of another P_3 and is then free, which is the obstruction of [10] and the reason the argument below needs whole components.

Call component j *reducible* if G_j has an optimal solution $S_j \subseteq \binom{W_j}{2}$ such that for all $u, v \in W_j$: (R1) if u, v lie in one cluster of $G_j \triangle S_j$, then $N_G(u) \setminus W_j = N_G(v) \setminus W_j$; (R2) otherwise $N_G(u) \cap N_G(v) \subseteq W_j$. This is Reduction Rule 6.2 of [13] for $H = G_j$.

Lemma 25 (Rule 6.2 at component level). *If component j is reducible with S_j, then $(G, \{G_i\}, k)$ is a YES instance iff $(G \triangle S_j, \{G_i\}_{i \ne j}, k - \tau_j)$ is. In the new instance the graphs G_i ($i \ne j$) are unchanged, still vertex-disjoint and induced, with the same τ_i; the excess ℓ' is unchanged; and for every $i \ne j$ the conditions (R1), (R2), hence reducibility, are unchanged. So one pass over the components, in any order, reaches an instance in which no component is reducible.*

Proof. The equivalence is Lemma 6.3 of [13] (cited; sound for every induced subgraph). Both k and $\sum\tau$ drop by τ_j. The set S_j consists of pairs with both ends in W_j, so no pair incident to a vertex of W_i, $i \neq j$, changes: $G[W_i]$ and all neighbourhoods $N(u)$, $u \in W_i$, are the same before and after, and (R1), (R2) for i refer only to these. □

Lemma 26 (Number of irreducible components)**.** *Let $(G, \{G_j\}_{j\le r}, k)$ be a YES instance with $\ell' = k - \sum_j \tau_j$ in which no component is reducible. Then $r \le 2\ell'$.*

Proof. Let S be an optimum of G, so $|S| \le k$. *Claim:* for every j, (a) $|S_j| \ge \tau_j + 1$, or (b) S contains a pair with exactly one end in W_j (an *external* pair of j). Suppose (a) fails; then $|S_j| = \tau_j$ by Lemma 24(4), so S_j is an optimal solution of G_j, and since j is not reducible, S_j violates (R1) or (R2). In $G \triangle S$ the induced subgraph on W_j is the cluster graph $G_j \triangle S_j$. Case (R1) violated: $u, v \in W_j$ are in one cluster of $G_j \triangle S_j$, so $uv \in E(G \triangle S)$ and, $G \triangle S$ being a cluster graph, $N_{G\triangle S}[u] = N_{G\triangle S}[v]$; but there is $y \notin W_j$ adjacent in G to exactly one of them, so one of uy, vy is in S: an external pair. Case (R2) violated: u, v are in different clusters of $G_j \triangle S_j$, hence non-adjacent in $G \triangle S$, and have a common G-neighbour $y \notin W_j$; if neither uy nor vy were in S, then $u\,y\,v$ would be an induced P_3 of $G \triangle S$. So again an external pair is in S.

Let J_a be the indices satisfying (a) and J_b the others, which satisfy (b). An external pair of j has exactly one end in W_j; it lies in no $\binom{W_i}{2}$ (for $i = j$ because of the end outside, for $i \neq j$ because $W_i \cap W_j = \emptyset$ and one end is in W_j), so it belongs to S_{ext}, and it is external for at most two indices (the components containing its ends). Hence $|S_{\text{ext}}| \ge |J_b|/2$. Since $|S| = \sum_j |S_j| + |S_{\text{ext}}|$ (disjoint pair sets), $|S| \ge \sum_{j\in J_a}(\tau_j + 1) + \sum_{j\in J_b}\tau_j + |J_b|/2 \ge \sum_j \tau_j + r/2$. With $|S| \le k$ this reads $\ell' \ge r/2$. □

The bound is on irreducible components, not on components with positive excess $\tau_j > p_j$: a reducible component may have large internal excess, and it is removed by Lemma 25 at no cost in ℓ'.

Proof of Theorem 22. Algorithm. (1) Compute the components of $I(H)$, the optima τ_j by the subset dynamic programme over W_j ($O(3^{|W_j|})$), and ℓ'; if $\ell' < 0$ answer NO (Lemma 24(4)). (2) For each component test reducibility and, if reducible, apply the rule (Lemma 25); the test of [13, Lemma 6.4] needs a constant number of exact cluster-editing computations on subgraphs of G_j plus polynomial time, which the subset programme provides in $O(3^{|W_j|})$ each. (3) Let r^* be the number of remaining components. If $r^* > 2\ell'$, answer NO (Lemma 26, contrapositive). Otherwise the current instance has budget $k^* = \sum_{\text{remaining}} \tau_j + \ell' \le r^* t_s + \ell' \le (2t_s + 1)\ell'$. Forget the packing and solve plain Cluster Editing with budget k^* by Böcker's algorithm in $O(1.62^{k^*} + n + m)$.

Correctness follows from Lemma 24 (the instance is of the required form and ℓ' is the excess), Lemma 25 (steps preserve the answer), and Lemma 26.

Time. The components have at least 3 vertices and are disjoint, so there are at most $n/3$ of them and steps (1), (2) take $O(n3^s) + n^{O(1)}$. Step (3) takes $1.62^{k^*} n^{O(1)}$ with $k^* \le (2t_s+1)\ell' \le (\binom{s}{2}+1)\ell$, since $2t_s \le \binom{s}{2}$ and $\ell' \le \ell$.

Kernel. After step (2) and the checks of step (3), apply the kernelization of Cluster Editing with at most $2k^*$ vertices (cited via [13]) to (G, k^*) and add the empty packing: an equivalent CEaMP instance with empty packing, budget $\le k^*$ and at most $2k^* \le (4t_s+2)\ell \le (s^2 - s + 2)\ell$ vertices. □

Remark 27 (Scope). Theorem 22 is FPT in (ℓ, s); it does not answer the main question, since s is not bounded by a function of w (a path or tree $I(H)$ has $w \le 1$ and arbitrarily large components), and the exponent $\binom{s}{2}\ell$ is of no practical interest. The argument uses neither pair-maximality nor Section 5. For a perfect matching it covers the smallest non-trivial instance of the case $c = 2$ left open in [10] for FPT in ℓ.

7 Structure of solutions with respect to a packing

This section collects counting lemmas that hold for every packing H (no maximality unless stated) and describe what a solution of size at most $|H| + \ell$ can look like. All proofs use only pair-disjointness, a budget, and counting; none uses optimality of the solution. The lemmas *do not* give an FPT algorithm in $\ell + w$ and no hardness; see Section 7.6 for what they do and do not explain, and Section 9.

Conventions. Here a pair is *hard* if it is not packed (in the language of the previous sections: *unpacked*); a *hard edge* is a hard pair in E and a *hard non-edge* a hard pair outside E. For a packing H let $c = c(H)$ be the maximum number of members of H through a vertex, $D = D(H) = \max_v |Q(v)|$ the maximum number of packed partners, and $w = \mathrm{tw}(I(H))$ (with $w = 0$ if $I(H)$ has no edge). A triple (G, H, ℓ) is a *YES instance* if $\mathrm{OPT}(G) \le |H| + \ell$. A *family* F of induced P_3's is *hard over* H if the sets $\mathrm{pairs}(P)$, $P \in F$, are pairwise disjoint and disjoint from the packed pairs of H; then $H \cup F$ is a packing with $|H| + |F|$ members.

Lemma 28 (Degrees)**.** *For every vertex v, $|Q(v)| = 2\,|\{P \in H : v \in P\}|$. Hence $D = 2c \le 2(w+1)$.*

Proof. Each member $P \ni v$ contributes the two packed pairs of P at v, i.e. two partners. Two members through v cannot have a common partner $x \neq v$, since they would then share the two vertices v, x, hence a pair. So the partner sets are disjoint and $|Q(v)| = 2|\{P \ni v\}|$. The inequality $c \le w + 1$ is stated in Section 2. □

7.1 Maximalisation and the hard-edge lemma

Lemma 29 (Hard families)**.** *Let H be any packing, (G, H, ℓ) a YES instance and F a family hard over H. Then $|F| \le \ell$. In general* $\mathrm{OPT} \ge |H| + |F|$.

Proof. $H \cup F$ is a modification-disjoint packing of $|H| + |F|$ induced P_3's. Lemma 1, applied to $H \cup F$ and an optimum solution, gives $\mathrm{OPT} \ge |H| + |F|$. In a YES instance $\mathrm{OPT} \le |H| + \ell$. □

Lemma 30 (MAX: maximalisation)**.** *Let H be any packing and M a maximal family hard over H (inclusion-maximal; it is found greedily in polynomial time). Put $H' = H \cup M$, $\ell' = \ell - |M|$. Then:*

(a) $\mathrm{OPT} \ge |H'|$. *In particular $|M| > \ell$ implies that (G, H, ℓ) is a NO instance.*

(b) $\mathrm{OPT} \le |H| + \ell$ *iff* ($|M| \le \ell$ *and* $\mathrm{OPT} \le |H'| + \ell'$). *So (G, H, ℓ) and (G, H', ℓ') are equivalent whenever $\ell' \ge 0$.*

(c) H' is a packing and is pair-maximal: no induced P_3 of G has all three pairs unpacked with respect to H'. ("Hard" now means hard with respect to H'; the pairs of M have become packed.)

(d) $\mathrm{tw}(I(H')) \le \mathrm{tw}(I(H)) + |M|$, *hence $w' \le w + |M| \le w + \ell$.*

(e) $c' \le \min\{c + |M|,\ w' + 1\}$ and $D' = 2c' \le D + 2|M|$.

A tree decomposition of $I(H')$ of width at most $w + |M|$ is obtained in polynomial time from one of $I(H)$.

Proof. (a) is Lemma 29 (first sentence: $\mathrm{OPT} \ge |H| + |M|$). (b) If $\mathrm{OPT} \le |H| + \ell$ then $|M| \le \ell$ by (a), and $|H'| + \ell' = |H| + \ell$. Conversely $|M| \le \ell$ and $\mathrm{OPT} \le |H'| + \ell' = |H| + \ell$. So (b) is arithmetic once (a) is known.

(c) The members of M have pairwise disjoint pair sets, disjoint from those of H, so H' is a packing. Suppose an induced P_3 Q has all its pairs hard with respect to H'. Then pairs(Q) is disjoint from pairs(H) and from pairs(P) for all $P \in M$, so $M \cup \{Q\}$ is a family hard over H ($Q \notin M$ because the pairs of Q are not in pairs(M)), contradicting maximality of M.

(d) Let $(T, \{B_t\})$ be a tree decomposition of $I(H)$ of width w (if $H = \emptyset$ take a single empty bag). The graph $I(H')$ has node set $H \cup M$ and $I(H')[H] = I(H)$. Put $B'_t = B_t \cup M$. Every node of H still occupies a connected subtree. Every node of M occupies all of T, which is connected. An edge of $I(H')$ with both ends in H is covered as before; an edge with an end $P \in M$ is covered by any bag containing its other end (every node of H lies in some bag; if both ends are in M every bag covers it). Bags have at most $w + 1 + |M|$ nodes, so the width is at most $w + |M|$.

(e) At most $|M|$ members of M pass through a vertex, so $c' \le c + |M|$; $c' \le w' + 1$ is Section 2 for H'. By Lemma 28, $D' = 2c' \le 2c + 2|M| = D + 2|M|$. □

The bound in (d) cannot be improved in general.

Example 31 (Tightness of (d))**.** Let G be the star with centre v and leaves $p, q, a_1, b_1, \ldots, a_m, b_m$, $H = \{(p, v, q)\}$ and $M = \{(a_i, v, b_i) : 1 \le i \le m\}$. The members of M are induced P_3's with pairwise disjoint pair sets, disjoint from pairs(H). M is maximal: every induced P_3 of a star is (x, v, y) with leaves x, y, and every pair xv is already packed. $I(H)$ is one node, tw $= 0$; $I(H \cup M) = K_{m+1}$, tw $= m = 0 + |M|$.

After applying Lemma 30, all the other lemmas of this section have to be applied to the primed instance (G, H', ℓ') with parameters $w' \le w + |M|$, $c' \le c + |M|$, $D' \le D + 2|M|$ and $\ell' = \ell - |M|$. This is the same normalisation as Remark 2, with the cost in treewidth made explicit.

Lemma 32 (HE: hard edges close up)**.** *Let H be pair-maximal and let ab, bc be hard edges (a, b, c distinct). Then $ac \in E$ or ac is packed.*

Proof. If $ac \notin E$ and ac were hard, then $a\,b\,c$ would be an induced P_3 with all three pairs unpacked, contradicting pair-maximality. So $ac \in E$ or ac is packed. □

No YES hypothesis is needed. Without pair-maximality the lemma fails ($H = \emptyset$ and any induced P_3). For an arbitrary packing H one applies it to $H' = H \cup M$ of Lemma 30(c), with "hard" meaning hard with respect to H'.

7.2 Hubs

Lemma 33 (HUB$^+$)**.** *Let H be any packing, (G, H, ℓ) a YES instance, u a vertex and $N_h(u) = \{x : ux \in E,\ ux \text{ not packed}\}$. Let Γ_u be the graph on $N_h(u)$ whose edges are the pairs $\{x, y\}$ with $xy \notin E$ and xy not packed (hard non-edges). Then:*

(a) $\nu(\Gamma_u) \le \ell$ (matching number);

(b) Γ_u has a vertex cover T with $|T| \le 2\ell$, and every pair of $N_h(u) \setminus T$ is an edge of G or is packed;

(c) $\alpha(G[N_h(u)]) \le 2\ell + c + 1$ where $c = c(H)$.

Proof. (a) Let $x_1y_1, \ldots, x_ty_t$ be a matching of Γ_u and $P_i = (x_i, u, y_i)$. Since $ux_i, uy_i \in E$ and $x_iy_i \notin E$, P_i is an induced P_3. Its three pairs ux_i, uy_i, x_iy_i are hard. The $2t$ vertices x_i, y_i are distinct and different from u, so the pairs ux_i, uy_i ($i \le t$) are pairwise distinct, and the pairs x_iy_i are pairwise distinct and do not contain u. So $\{P_1, \ldots, P_t\}$ is hard over H and $t \le \ell$ by Lemma 29.

(b) The vertex set of a maximum matching of Γ_u is a vertex cover (by maximality) of size $2\nu \le 2\ell$. A pair of $N_h(u) \setminus T$ is not an edge of Γ_u, i.e. it is an edge of G or packed.

(c) Let J be independent in $G[N_h(u)]$. By (b), every pair of $J \setminus T$ is a non-edge of G and not an edge of Γ_u, hence packed. Put $J' = J \setminus T$. A packed pair of J' lies in a member P of H and is a non-edge, so it is the pair of the two ends of P (the other two pairs of P are edges); in particular $|V(P) \cap J'| \le 2$. If $J' \neq \emptyset$ fix $x \in J'$; every $y \in J' \setminus \{x\}$ gives a member $P_y \ni x, y$, and $P_y \neq P_{y'}$ for $y \neq y'$ because $|V(P_y) \cap J'| \le 2$. So $|J'| - 1 \le c$. Hence $|J| \le |T| + |J'| \le 2\ell + c + 1$. □

Corollary 34 (HUB, hard-independent sets)**.** *In the setting of Lemma 33, if $X \subseteq N_h(u)$ and every pair of X is a hard non-edge, then $|X| \le 2\ell + 1$.*

Proof. X is a clique of Γ_u, so it contains a matching of size $\lfloor |X|/2 \rfloor$, and $\lfloor |X|/2 \rfloor \le \ell$ by Lemma 33(a). □

Remark 35 (HUB$^+$ versus an independence-number bound). (1) Bounding $\alpha(G[N_h(u)])$ through the corollary (greedy choice of a subset of an independent set that is also free of packed pairs) gives only $(c+1)(2\ell+1) = (D/2+1)(2\ell+1)$, which is weaker than (c). (2) The matching form (a) is the relevant statement for Vertex Cover on the non-adjacency graph of $N(u)$: let $N_h(u)$ be two disjoint cliques of G of size m with no packed pair between them. Then $\alpha = 2$, but (a_i, u, b_i), a_i, b_i from different cliques, are m pair-disjoint induced P_3's, and $m \le \ell$ in a YES instance. Large independent sets are not the obstruction; large matchings of Γ_u are. (3) For a general H the lemma is applied to H' of Lemma 30 with c', ℓ'.

7.3 Pieces and small merges

Fix a packing H and a solution S ($G \triangle S$ is a cluster graph) with $|S| \le |H| + \ell$. Let $U = S \setminus \bigcup_i \text{pairs}(P_i)$ be the unpacked part and $u = |U|$; by Lemma 1, $u \le |S| - |H| \le \ell$. Put $F = U \cap E$ (deleted hard edges) and $I = U \setminus E$ (inserted hard pairs). The *pieces of S* are the connected components of the graph (V, K), $K = \{\text{hard edges of } G\} \setminus F$ (kept hard edges); isolated vertices are singleton pieces. *Pieces depend on S, not only on (G, H).* Every edge of K is an edge of $G \triangle S$, so every piece lies inside one cluster of $G \triangle S$.

Lemma 36 (SM: small merge)**.** *Let S, u, I, pieces as above, $D \ge \max_v |Q(v)|$ (e.g. $D = D(H)$), and let C be a cluster of $G \triangle S$.*

(i) If $A \neq B$ are pieces contained in C, then every pair ab ($a \in A, b \in B$) is packed or belongs to I.

(ii) For such A, B with $|A| \ge |B|$: $|B|\,(|A| - D) \le |I \cap (A \times B)| \le u$, hence $|A| \le D + u \le D + \ell$.

(iii) If C is the union of at least two pieces, then $|C| \le \max\{D + u + 1,\ 2D,\ 2u\}$.

Proof. (i) Since $a, b \in C$, ab is an edge of $G \triangle S$, i.e. $ab \in E$ xor $ab \in S$. Suppose ab is hard. If $ab \in E$ then $ab \notin S$; as $F \subseteq S$, $ab \notin F$, so $ab \in K$ and a, b lie in one piece, a contradiction. So $ab \notin E$, hence $ab \in S$, and as ab is hard, $ab \in U \setminus E = I$.

(ii) Each vertex of B has at most D packed partners, so at most $D|B|$ pairs of $A \times B$ are packed. By (i) all other pairs of $A \times B$, at least $|A||B| - D|B|$, lie in I, and $|I| \le |U| = u$. If $|A| \ge D + u + 1$ then $|B|(|A| - D) \ge u + 1$ because $|B| \ge 1$, a contradiction; so $|A| \le D + u$.

(iii) Let A be a largest piece in C, $a = |A|$ and $r = |C| - a \ge 1$. By (i), every pair $\{x, y\}$ with $x, y \in C$ in different pieces is packed or in I.

Case $a > D$. Each $x \in C \setminus A$ has at most D packed partners, so at least $a - D \geq 1$ vertices $y \in A$ give a pair xy in I. These pairs are distinct for distinct (x, y), so $r(a - D) \leq |I| \leq u$. Then $r \leq u$ and $a \leq D + u/r$, so $|C| = a + r \leq D + u/r + r$. The function $r \mapsto r + u/r$ is convex on $[1, u]$ and takes the value $u + 1$ at both ends, so $|C| \leq D + u + 1$. (The case does not occur if $u = 0$.)

Case $a \leq D$. All pieces of C have at most D vertices. Each $x \in C$ has at least $|C| - D$ vertices of C outside its piece, at most D of which are packed partners; so x lies in at least $|C| - 2D$ pairs $\{x, y\}$ of I with $y \in C$. Summing over x counts each pair twice: $2|I| \geq |C|(|C| - 2D)$. If $|C| > 2D$, then $|C| - 2D \geq 1$ and $|C| \leq 2|I| \leq 2u$. So $|C| \leq \max\{2D, 2u\}$. □

Remark 37 (Qualifiers and relation to the original bound). (1) The qualifier "pieces sharing a cluster" is essential: a single piece may be an entire cluster of unbounded size, and a statement "pieces have size at most $D+\ell$" is false. (2) The bound (iii) implies the weaker bound $|C| \leq \max\{2D+\ell, 2\ell\}$ with which the lemma first appeared in our notes: $2u \leq 2\ell$ and $2D \leq 2D+\ell$ are clear; $D+u+1 \leq 2D+\ell$ if $D \geq 1$; if $D = 0$ and $u \geq 1$ then $u + 1 \leq 2u \leq 2\ell$; and if $D = u = 0$ no cluster with two pieces exists (the two cases of the proof are both impossible). (3) The bound (iii) is attained in some small instances of our tests (slack 0), for instance two singleton pieces and one inserted pair; we do not claim it is tight in general. (4) Lemma 36 says nothing about the *degrees* in a graph of pieces; it bounds the sizes of pieces that share a cluster and the number of cross pairs that are not packed.

7.4 The soft graph has bounded treewidth

Let $W = (V, \{\text{packed pairs}\})$ be the *soft graph*.

Proposition 38 (Treewidth of the soft graph)**.** $\mathrm{tw}(W) \leq 3(w + 1) - 1 = 3w + 2$ *if $H \neq \emptyset$; W is edgeless if $H = \emptyset$. The bound is attained for $w = 0$ (H a single P_3, W a triangle).*

Proof. Let $(T, \{B_t\})$ be a tree decomposition of $I(H)$ of width w and $T_P = \{t : P \in B_t\}$, a subtree of T for each $P \in H$. Put $B'_t = \bigcup_{P \in B_t} V(P)$, so $|B'_t| \leq 3(w + 1)$. For every vertex $x \notin V(H)$ add a new leaf bag $\{x\}$ to an arbitrary node. *Edges:* a packed pair lies in $V(P)$ for some $P \in H$, and $P \in B_t$ for some t, so the pair lies in B'_t. *Connectivity:* for $x \in V(H)$ the nodes t with $x \in B'_t$ form $\bigcup_{P \ni x} T_P$. The members through x pairwise share x, so they are pairwise adjacent in $I(H)$ and, the edge being covered by a bag, their subtrees T_P pairwise intersect. A union of connected subtrees that pairwise intersect is connected (two points of the union lie in subtrees $T_P, T_{P'}$ that meet; join them through a common node). The new leaf bags are single bags of size 1. So this is a tree decomposition of W of width at most $3(w + 1) - 1$. □

Computationally, for $w \geq 1$ the largest value of $\mathrm{tw}(W)$ that we saw on small instances is much smaller than $3w + 2$; we do not claim the bound is tight beyond $w = 0$.

7.5 Computer verification

All statements of this section were checked by brute force with code written independently of the other scripts (`code/w5_packing_lemmas.py`, all set partitions, $n \leq 8$, 4000 random graphs: random, noisy clusters, hubs; for each a random partial packing H and its maximalisation $H \cup M$): Lemma 28 ($D = 2c$), Lemma 30(a)–(e), Example 31 for $m \leq 5$, Lemma 32, Lemma 33 and Corollary 34 for every vertex and every ℓ with $\mathrm{OPT} \leq |H| + \ell \leq |H| + 3$, Lemma 36 on every solution of cost at most $|H| + 3$ (about $5.7 \cdot 10^5$ clusters with two or more pieces, including slack 0 in (iii) on some cases), and Proposition 38 (exact treewidth). No violation; the variants $\alpha \leq 2\ell + c$ and $|C| \leq \max\{D + u, 2D, 2u\}$ are falsified by the same script, so the checks are not vacuous. This is evidence for the statements; the proofs above are the justification.

7.6 Why simple hardness encodings fail

This subsection is a heuristic discussion, not a theorem. It explains why three natural ways of building W[1]-hardness for $\ell + w$ do not work as stated; it does not exclude other gadgets and it does not prove fixed-parameter tractability, which remains open (Open Problem 1). A reduction from a problem with parameter k must keep $\ell + w \leq g(k)$; grids of side k inside $I(H)$ are therefore allowed, but structures that grow with the instance size are not.

Hub of deletions. A hub u is adjacent to vertices $x_1, \ldots, x_m$ and the instance asks to keep a subset of $N(u)$ forming a clique and delete the rest, i.e. Vertex Cover on the non-adjacency graph of $N(u)$. Lemma 33 shows what survives in a YES instance: the graph of *hard* non-adjacent pairs inside $N_h(u)$ has a matching of size at most ℓ and a vertex cover of size at most 2ℓ, so at most 2ℓ hub neighbours carry hard conflicts; the other non-adjacencies are adjacencies or packed pairs, and a vertex has at most $2c$ of the latter. The relevant obstruction is a large matching (Remark 35(2)), not a large independent set, and any such matching immediately forces a NO answer. Non-adjacencies realised by packed pairs are local, with degree at most $2c \leq 2(w+1)$ in the soft graph. This is why Vertex Cover through a hub cannot encode an unbounded source graph; it does not address hubs whose conflicts are realised by packed pairs; a vertex has at most $2c$ of these, so many of them force large c, hence large w.

Hub of insertions. A hub u with classes $V_1, \ldots, V_k$ chooses, by inserting uv_{i,j_i}, one vertex per class; the selected vertices land in one cluster C and compatibility is encoded by adjacency of the chosen vertices. By Lemma 36(i), two selected vertices in different pieces of C form a packed pair or an inserted pair (at most ℓ of these); adjacency by a hard edge that is not deleted puts them in the same piece, and Lemma 36(ii) bounds the size of merged pieces by $D + \ell$. So the compatibility relation between selected vertices in different pieces can only be realised by packed pairs, which form the soft graph, of treewidth at most $3w + 2$ (Proposition 38), or by a bounded number of inserted pairs. The caveats: Lemma 36 bounds sizes of pieces, not the degree of a compatibility graph; at most ℓ hard edges may be deleted, which a gadget could exploit; and pieces depend on the chosen solution. The argument shows that the hard graph cannot serve as a channel for $[n]$-valued information between selected vertices, nothing more.

Grid tiling. A tile vertex t_{ab} is joined by hard edges to a row vertex x_a and a column vertex y_b ($x_a y_b$ a non-edge). The triples (x_a, t_{ab}, y_b) are induced P_3's; if their pairs are hard they are pairwise pair-disjoint (their pairs are $x_a t_{ab}$, $t_{ab} y_b$, $x_a y_b$, all different for different (a, b)), so by Lemma 29 at most ℓ of them can be hard, whereas a grid-tiling encoding needs about n^2 tiles and n is not bounded by $g(k)$. So all but at most ℓ of them are packed; then some row vertex x_a lies in at least $n - \ell$ members (if every row has n tiles), $c \geq n - \ell$, hence $w \geq n - \ell - 1$, again unbounded in k. Mixed and replicated gadgets (several copies of row and column vertices, tiles realised by chains) were not excluded by a proof; the examination in our notes found no variant that carries information on $[n]$ across a tree decomposition of bounded width, which is evidence only.

What remains open. The lemmas above make every "simple" source of unbounded information in $I(H)$-width-bounded instances expensive. They leave open whether a bounded set of candidate positions exists for the inserted pairs whose endpoints are at soft distance 2 from $V(H)$ and lie in tiny pieces (our notes call this the locality statement for tiny pointers); this statement is neither proved nor disproved here, and the FPT question for $\ell + w$ stays an open problem.

8 Limits of the parameterization

8.1 An ETH lower bound in the exponent

Theorem 39. *Assume ETH. There is no algorithm that decides CEaMP with $\ell = 0$ in time $2^{o(\mathrm{tw}(I(H)))}n^{O(1)}$. This holds even when every vertex lies in at most 23 packed P_3's (so $I(H)$ has maximum degree at most 66), and whether or not a tree decomposition is supplied.*

Proof. Li, Pilipczuk and Sorge [10, Thm. 1] give a polynomial reduction from 3-SAT to CEaMP with $\ell = 0$ and at most 23 packed P_3's per vertex. We use it as a black box together with its size accounting. After a linear preprocessing every clause has exactly three distinct variables and every variable occurs at least twice; let N be the number of variables and m the number of clauses, so $\sum_i m_i = 3m$ for the occurrence counts m_i. By the gadget descriptions of [10] (read in the audit, not recomputed constant by constant): a variable gadget has $4m_i$ cliques of 5 vertices and 25 packed P_3's per triple of consecutive cliques, hence $20m_i$ vertices and $50m_i$ packed P_3's; a clause gadget has $20 + 114$ vertices (clique sizes $1, 1, 4, 14$ and $34, 46, 34$), $6 + 12$ transferring P_3's, and $O(1)$ padding P_3's from a triangle packing on cliques of constant size. Hence $|V| \approx 194m$ and $|H| = O(m)$ with an absolute constant (not computed; a crude count gives about $3000m$). By the Sparsification Lemma (Impagliazzo, Paturi and Zane [7]; see also [4]) we may assume $m = O(N)$ at the cost of $2^{o(N)}$ extra time. Then $|H| = O(N)$ and $\mathrm{tw}(I(H)) \le |H| - 1 = O(N)$ (one bag). An algorithm with running time $2^{o(\mathrm{tw}(I(H)))}n^{O(1)}$, which may compute its own decomposition, would therefore decide 3-SAT in time $2^{o(N)}$, contradicting ETH. □

Remark 40 (Caveats). (1) The bound is on the exponent only: it does not show that the base 3 (or 7) is tight, and $2^{o(w)}$ excludes $3^{o(w)}$ and $7^{o(w)}$ alike. A SETH-type bound $(3 - \varepsilon)^w$ is open. (2) The proof uses only $w \le |H| - 1$; it does not exhibit hardness for instances of small $|H|$ relative to w. (3) The constants of the size table were read from the reduction and not re-derived one by one; the structural claim (every gadget has $O(1)$ size per clause or occurrence) was re-derived. (4) The statement refers to the version-3 arXiv text of [10].

8.2 Other parameters do not suffice

Proposition 41. *CEaMP is NP-hard when $I(H)$ has no edge ($w = 0$, vertex-disjoint packing), with ℓ part of the input. Hence w alone is not enough, and (by Theorem 1 of [10]) neither is $\ell = 0$ alone.*

Proof. Cluster Editing is NP-hard (e.g. already on graphs of maximum degree 6 [8]). Given (G, k) compute greedily a maximal set H of vertex-disjoint induced P_3's (if none, G is a cluster graph). Vertex-disjoint implies modification-disjoint and $I(H)$ is edgeless. Since $\mathrm{OPT} \ge |H|$, (G, k) is a YES instance iff $(G, H, k - |H|)$ is (a NO instance if $k < |H|$). □

Proposition 42. $\mathrm{tw}(G)$ *and* $\mathrm{tw}(I(H))$ *are incomparable, in the following sense. (a) For the star $G = K_{1,2m}$ ($m \ge 1$) every packing $H \ne \emptyset$ has $I(H) = K_{|H|}$ and every YES instance (G, H, ℓ) satisfies $\ell + \mathrm{tw}(I(H)) \ge 2m - 2$, while $\mathrm{tw}(G) = 1$. (b) For the $k \times k$ grid G and a family H of vertex-disjoint induced P_3's along rows, $\mathrm{tw}(I(H)) = 0$ and $\mathrm{tw}(G) = k$.*

Proof. (a) Leaves have degree one, so the centre of any induced P_3 is the hub v; all packed P_3's are (a, v, b) with distinct leaves (the pairs va are disjoint), pairwise sharing v, so $I(H) = K_p$, $p = |H| \le m$, $\mathrm{tw} = p - 1$. If the hub's cluster contains j leaves, the cost is $(2m - j) + \binom{j}{2}$ (the other leaves are best left as singletons), which is minimized, with value $2m - 1$, at $j \in \{1, 2\}$; so $\mathrm{OPT} = 2m - 1$ and a YES instance has $\ell \ge 2m - 1 - p$, i.e. $\ell + \mathrm{tw}(I(H)) \ge 2m - 2$. (b) Rows of a grid are induced P_3's (no triangles) and $\mathrm{tw}(k \times k \text{ grid}) = k$ is standard. □

Thus hubs do not make $\mathrm{tw}(I(H))$ large by themselves; they force a tradeoff: small $\mathrm{tw}(I(H))$ on a star forces large ℓ.

9 Open problems

Notation as above; $w = \mathrm{tw}(I(H))$.

Open Problem 1 (Main question: (ℓ, w) in general)**.** Is CEaMP fixed-parameter tractable parameterized by $\ell + w$, i.e. solvable in $f(\ell, w)n^{O(1)}$ time, or is it W[1]-hard in $\ell + w$? Known: XP in ℓ and FPT in w for each fixed ℓ only via $n^{2\ell}$ (Theorem 6); FPT in ℓ when $w = 0$ by [13], and FPT in (ℓ, s) when the components of $I(H)$ span at most s vertices (Theorem 22); the case $H = \emptyset$ (plain Cluster Editing, $\ell = k$) is FPT. Note that adding c to the parameter changes nothing, since $c \leq w + 1$. A positive answer needs a bound $g(\ell, w)$ on the number of candidate positions for each unpacked edit (Lemma 11 gives none; Theorem 14 and Proposition 19 only organise the edits touching R and do not bound the HH-edits); the lemmas of Section 7 (MAX, HUB$^+$, SM) bound hubs, merged pieces and the soft graph but do not give such a bound for inserted pairs between tiny pieces; a negative answer has to encode $[n]$-valued information through unpacked edits.

Open Problem 2 ($w = 1$ with unbounded tree components)**.** If $I(H)$ is a forest ($w \leq 1$, hence $c \leq 2$) with components of unbounded size, is CEaMP FPT in ℓ? Theorem 22 settles bounded components, in particular perfect matchings (Corollary 23), but its counting argument is at the level of whole components and gives nothing for a single large tree component.

Open Problem 3 (Half-integral and intermediate packings)**.** Is CEaMP with $c = 2$ FPT in ℓ? What is the complexity of $\ell = 0$ for $3 \leq c \leq 22$? Both are open in [10], whose authors say a W[1]-hardness result for $c = 2$ "would also not be surprising".

Open Problem 4 (Base of the exponent)**.** For $\ell = 0$, does a $(3 - \varepsilon)^w n^{O(1)}$ (or even $2^{o(w)}$ with a smaller base) algorithm exist, or is there a SETH-type lower bound? Theorem 39 gives only the exponent.

Open Problem 5 (Treewidth of G)**.** What is the complexity of CEaMP with $\ell = 0$ (or of Cluster Editing) when $\mathrm{tw}(G)$ is bounded? We found no published result on Cluster Editing parameterized by treewidth alone (Section 10), so this must be proved rather than cited.

10 Related work

Packing lower bounds. van Bevern, Froese and Komusiewicz [13] parameterize F-free editing above the size of a vertex-disjoint packing; for Cluster Editing they obtain an $O(4^\ell \ell^3 + m + n)$-time algorithm for P_3 packings (Thm. 6.13), and for cost-t packings an $O(1.62^{(2t+1)\ell} + nm + n1.62^t)$ algorithm and a kernel with $O(t\ell)$ vertices. They show NP-hardness at $\ell = 0$ of K_3-free editing for edge-disjoint packings and of K_q-free editing, $q \geq 6$, for vertex-disjoint packings, and leave open Cluster Editing above edge-disjoint packings and the stronger modification-disjoint bound. Li, Pilipczuk and Sorge [10] answer the latter: NP-hard at $\ell = 0$ with at most 23 packed P_3's per vertex, $n^{2\ell+O(1)}$ for $c = 2$, and open for $c = 2$ in ℓ and for $3 \leq c \leq 22$ at $\ell = 0$. They report "grid-like structures" as the obstacle. Our $\mathrm{tw}(I(H))$ parameter does not appear in either paper.

Twins and normal forms. The twin lemma (Lemma 10) is the critical-clique property behind the kernelizations of Guo [6]; the closed-neighbourhood reduction rule that generalises it is Reduction Rule 6.2 of [13], which Section 6 applies to components of $I(H)$. Pair-maximality and the structure of R (Lemma 12) are our own bookkeeping; we make no priority claim.

Citations and treewidth. Of the citing works recorded by the notes, we opened the survey of Crespelle et al. [3], the vertex-splitting paper [12] (it uses modification-disjoint P_3 packings as a lower-bound tool), Bocci et al. [1] and Garvardt et al. [5]; none improves $c = 2$, settles $3 \le c \le 22$, or uses a treewidth-type parameter. Two further citing items (the PACE 2021 report and a 2021 thesis) were not opened, and citation coverage was incomplete (Semantic Scholar listed seven records, OpenAlex a count of two); this is a "not found", not proof of absence. We found no paper stating the complexity of plain Cluster Editing or Deletion parameterized by treewidth. The nearby s-Club Cluster Edge Deletion is FPT in $s + \mathrm{tw}$ but W[1]-hard in pathwidth [11]; as a deletion problem with a diameter bound it does not transfer to editing. Komusiewicz and Uhlmann [8] show that Cluster Editing is NP-hard at maximum degree 6, with no $2^{o(k)}$ algorithm under ETH even at bounded degree; this is consistent with the fact that bounded degree of $I(H)$ does not give tractability (Theorem 39).

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